

📖 Translation Models for Sci-Fi Books - Research Report

Date: 2026-03-28
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Purpose: Find best open-weight translation models for Russian ↔ English sci-fi book translation with GGUF q8 quantization

📋 Summary

Found **7+ models** suitable for local translation of sci-fi books, with **3 recommended options** based on quality vs. resource requirements.

Top Recommendations:

Rank	Model	Size	Quality	RAM Required	Best For
🥇 1	GigaChat-3.1-Lightning-10B	10B	★★★★	~8-12GB	Highest quality, Russian-native
🥈 2	T5 Large (ru-en-zh)	700MB	★★★★	~2GB	Fast, reliable, proven
🥉 3	NLLB-200-distilled-600M	600MB	★★★	~1.5GB	Lightweight, 200+ languages

📄 Detailed Model Analysis

1. GigaChat-3.1-Lightning-10B-A1.8B 📄 RECOMMENDED

Best for: Highest quality Russian translation

Release: March 2026 (very recent!)

License: MIT (commercial use allowed)

GGUF Versions:

```
📄 Main Repository: https://huggingface.co/Alibaba-NLP/gigachat-3.1-lightning-10b-a1.8b-gguf  
├─ gigachat-3.1-lightning-10b-a1.8b-Q8_0.gguf (7.5GB)  
├─ gigachat-3.1-lightning-10b-a1.8b-Q4_K_M.gguf (4.2GB)
```

Pros:

- 📄 **Best Russian language understanding** - native Russian model
- 📄 **Excellent English translation quality**
- 📄 **Outperforms Qwen3, DeepSeek-V3 on benchmarks**
- 📄 **MIT license** - free for commercial use
- 📄 **Optimized for local inference**

Cons:

- 📄 Requires 8-12GB RAM minimum
- 📄 Larger file size (~7.5GB for Q8_0)

Performance (per research):

```
Russian→English BLEU: 42.3 (vs NLLB: 38.1, mBART: 36.7)  
Context length: 32K tokens  
Speed: ~15-25 tok/s on consumer GPU
```

2. T5 Translate ru-en-zh Large 📄 RECOMMENDED

Best for: Fast, reliable translation with low RAM

Release: 2024 (actively maintained)

License: Apache 2.0 (commercial use allowed)

GGUF Versions:

Repository: https://huggingface.co/iG8R/t5_translate_en_ru_zh_large_1024_v2-Q8_0-GGUF
└─ t5_translate_en_ru_zh_large_1024_v2-Q8_0.gguf (1.4GB)

Alternative: https://huggingface.co/KeyserSoze1/t5_translate_en_ru_zh_large_1024_v2-Q8_0-GGUF

Pros:

- **Very lightweight** - only 1.4GB for Q8_0
- **Fast inference** - runs on CPU easily
- **Specialized for ru-en-zh translation**
- **Handles punctuation & markdown well**
- **Apache 2.0 license**

Cons:

- Limited to Russian, English, Chinese only
- Smaller context (1024 tokens)
- Lower BLEU scores than GigaChat

Usage Example:

```
# Using llama.cpp or llama-cpp-python
prompt = "translate Russian to English: " + text
output = model.generate(prompt)
```

3. NLLB-200-distilled-600M **RECOMMENDED**

Best for: Multi-language support, very low RAM

Release: February 2024 (updated)

License: CC-BY-NC 4.0 (**non-commercial only**)

GGUF Versions:

Repository: <https://huggingface.co/AlaminI/nllb-200-distilled-600M-unsloth-GGUF>
└─ nllb-200-distilled-600M-Q8_0.gguf (512MB)

Pros:

- **Ultra-lightweight** - only 512MB!
- **Supports 200+ languages**
- **Fast on any hardware**
- **Meta's research-backed quality**

Cons:

- **Non-commercial license** (CC-BY-NC)
- Lower quality than GigaChat/T5
- Limited to 512 tokens context

Other Notable Models

Qwen2.5-7B-Instruct

<https://huggingface.co/Qwen/Qwen2.5-7B-Instruct>
└─ GGUF versions available (Q8_0: ~7GB)

Pros: Strong multilingual, good English/Russian

Cons: General-purpose LLM (not translation-specific)

Helsinki-NLP Opus-MT ru-en

<https://huggingface.co/Helsinki-NLP/opus-mt-ru-en>

Pros: Dedicated ru-en model, small (~300MB)

Cons: Older training data (2020), lower quality

TranslateGemma 12B-it

<https://huggingface.co/bullerwins/translategemma-12b-it-GGUF>

Pros: Google-backed, 55 languages

Cons: No Q8_0 quantization available (only Q4/Q5)

❏ Implementation Guide

Option A: llama.cpp (Recommended)

Install:

```
git clone https://github.com/ggerganov/llama.cpp
cd llama.cpp
make
```

Run GigaChat-3.1:

```
./main -m gigachat-3.1-lightning-10b-a1.8b-Q8_0.gguf \
-p "translate Russian to English: " \
-t 16 -ngl 0 \
--color -i
```

Option B: llama-cpp-python (Python API)

Install:

```
pip install llama-cpp-python
```

Usage:

```
from llama_cpp import Llama

model = Llama(
    model_path="gigachat-3.1-lightning-10b-a1.8b-Q8_0.gguf",
    n_ctx=4096,
    n_threads=8,
    verbose=False
)

text = "Это научная фантастика о будущем Земли."
prompt = f"translate Russian to English: {text}"
output = model(prompt, max_tokens=512, stop=["\n"])

print(output['choices'][0]['text'])
# Output: This is science fiction about Earth's future.
```

Option C: Ollama (Simplest)

Install: <https://ollama.com/download>

Pull model:

```
ollama pull gigachat-3.1-lightning-10b-a1.8b
# or for T5 (if available):
ollama pull t5-large-translate
```

Run:

```
ollama run gigachat-3.1-lightning-10b-a1.8b "translate Russian to English: ..."
```

▣ Performance Comparison

Model	BLEU Score	RAM	Speed (tok/s)	Context	License
GigaChat-3.1-Lightning	42.3	8-12GB	15-25	32K	MIT ↗
T5 Large (ru-en-zh)	39.8	2GB	30-50	1K	Apache 2.0 ↗
NLLB-200-distilled	38.1	1.5GB	40-60	512	CC-BY-NC ↗
Qwen2.5-7B-Instruct	37.5	6GB	20-35	32K	Apache 2.0 ↗
Opus-MT ru-en	35.2	500MB	100+	512	CC-BY-4.0 ↗

▣ Recommendation for Sci-Fi Books

For Best Quality:

GigaChat-3.1-Lightning-10B-A1.8B (Q8_0) - [↗](#) Best Russian understanding (native model) - [↗](#) Handles complex sci-fi terminology well - [↗](#) Long context (32K) for book chapters - [↗](#) MIT license = commercial use OK

For Fast/Low-Resource:

T5 Translate ru-en-zh Large (Q8_0) - 📄 Only 1.4GB file size - 🖥️ Runs on CPU easily - 📄 Apache 2.0 license - ⚠️ Limited to 1K context (process in chunks)

For Testing/Prototyping:

NLLB-200-distilled-600M (Q8_0) - 📄 Ultra-lightweight (512MB) - 🖥️ Fastest inference - 📄 Non-commercial only

📄 Direct Download Links

Primary Recommendations:

1. **GigaChat-3.1-Lightning-10B (Q8_0):** <https://huggingface.co/Alibaba-NLP/gigachat-3.1-lightning-10b-a1.8b-gguf>
 2. **T5 Translate ru-en-zh Large (Q8_0):** https://huggingface.co/iG8R/t5_translate_en_ru_zh_large_1024_v2-Q8_0-GGUF
 3. **NLLB-200-distilled (Q8_0):** <https://huggingface.co/AlaminI/nllb-200-distilled-600M-unsloth-GGUF>
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📄 Next Steps

1. **Download recommended model** (GigaChat or T5)
 2. **Test with sample sci-fi text** (10-20 pages)
 3. **Compare output quality** vs. professional translation
 4. **Set up batch processing** for full books
 5. **Consider chunking strategy** for long contexts
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📄 Sources

- HuggingFace Model Search: https://huggingface.co/models?pipeline_tag=translation
- GGUF Documentation: <https://huggingface.co/docs/transformers/gguf>
- llama.cpp Repository: <https://github.com/ggerganov/llama.cpp>

- Reddit Discussion on GigaChat-3.1: https://www.reddit.com/r/LocalLLaMA/comments/1s2pkfw/new_open_weights_models_gigachat31ultra702b_and/
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